AI COMMUNICATION SYSTEMS STUDY KIT

VERIFICATION REPORT

A checklist and evidence template to verify the AM/FM simulation project before submission.

Student name: _____

Department: Electronics and Communication Engineering (ECE)

College: _____

Date tested: _____

Important: This document is a verification template, not a certification that the project has passed. Fill in the status and evidence after you actually run and inspect the files.

1. Verification Scope

This report template covers the Python simulation workflow, the message and carrier reference signals, AM and FM waveform generation, saved graph files, documentation, and final submission links.

2. Test Checklist

ID	Test	Expected result	Status / evidence
T01	Run simulation code	No errors; execution completes	■ Pass ■ Fail ■ Not tested
T02	Message signal	Expected low-frequency waveform is plotted	■ Pass ■ Fail ■ Not tested
T03	Carrier signal	Higher-frequency sinusoidal carrier is plotted	■ Pass ■ Fail ■ Not tested
T04	AM waveform	Amplitude envelope follows the message; chosen modulation index is documented	■ Pass ■ Fail ■ Not tested
T05	FM waveform	Spacing changes with message; ideal amplitude stays constant	■ Pass ■ Fail ■ Not tested
T06	Graph export	All expected PNG files exist and open	■ Pass ■ Fail ■ Not tested
T07	Report and presentation	Files open; names, diagrams, and explanations are checked	■ Pass ■ Fail ■ Not tested
T08	Prompt library / automation guide	Documents open and match the implemented project	■ Pass ■ Fail ■ Not tested
T09	GitHub repository	Link opens; uploaded files are visible to intended audience	■ Pass ■ Fail ■ Not tested

3. Results Summary

Total tests: 9 Passed: _____ Failed: _____ Not tested: _____

Overall result (select after testing): ■ Pass ■ Pass with issues ■ Fail ■ Not yet verified

4. Evidence to Attach

■ Screenshot of the successful code execution or notebook output.

■ The actual message, carrier, AM, and FM graph images.

■ A file listing or screenshot showing that the graphs were saved.

■ The final report and presentation PDFs.

■ The working GitHub repository URL, if created.

5. Issues Found and Fixes

Issue 1: _____
Fix / action: _____

Issue 2: _____
Fix / action: _____

6. Limitations and Notes

A visual check alone does not prove that equations or code are fully correct. Verify the equations, simulation parameters, and plotted behavior. A Python script that calculates signals and saves plots is automation; do not label it as an AI model unless a real AI component has been implemented and tested.

7. Final Sign-off

Tested by: _____
Date: _____
Signature (if required): _____

This is a fill-in template. Do not report tests as passed unless you performed them and observed the expected result.